SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I Shaik Thaheer-192425347 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: *sk.Thaheer*

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

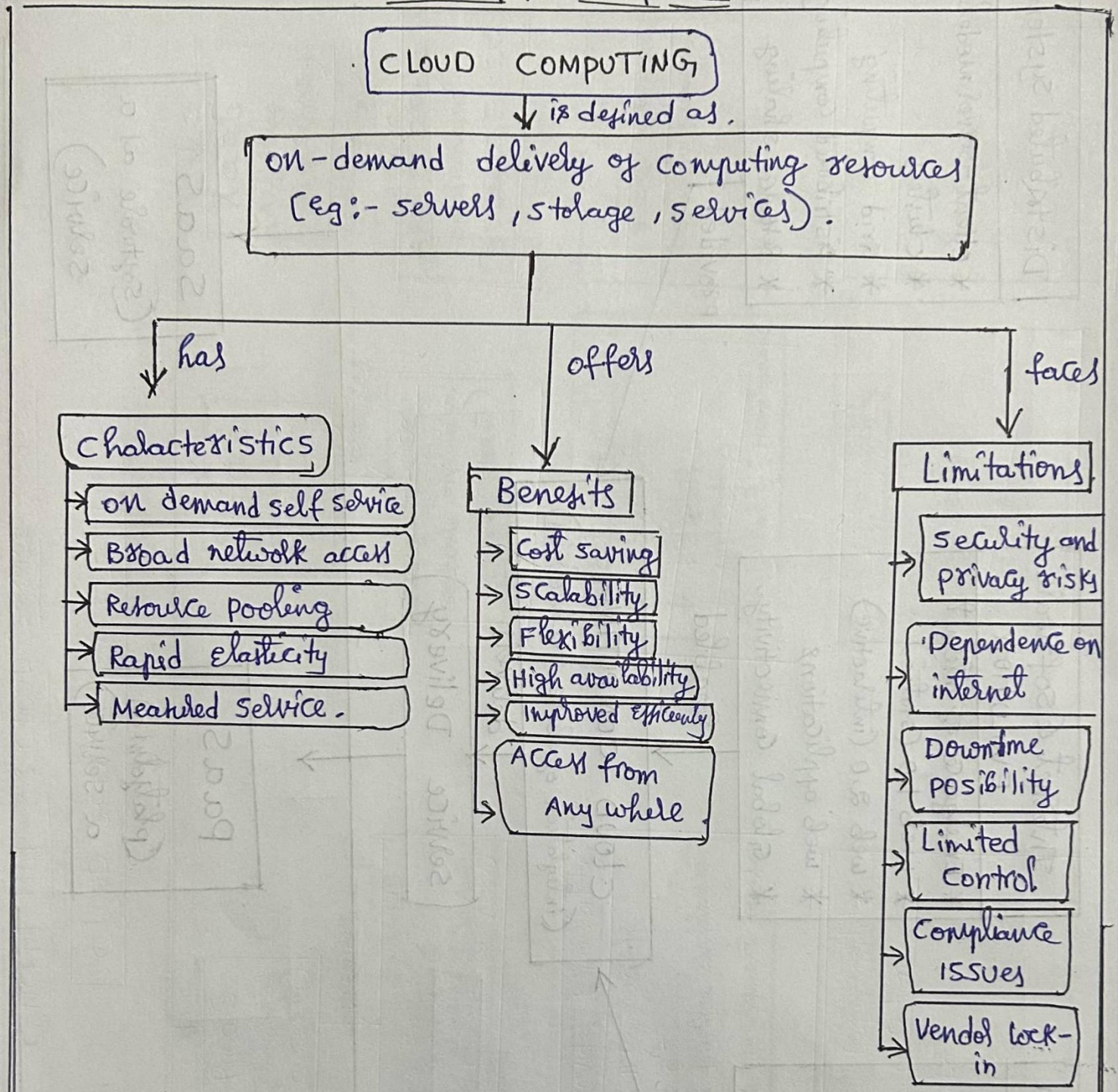
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

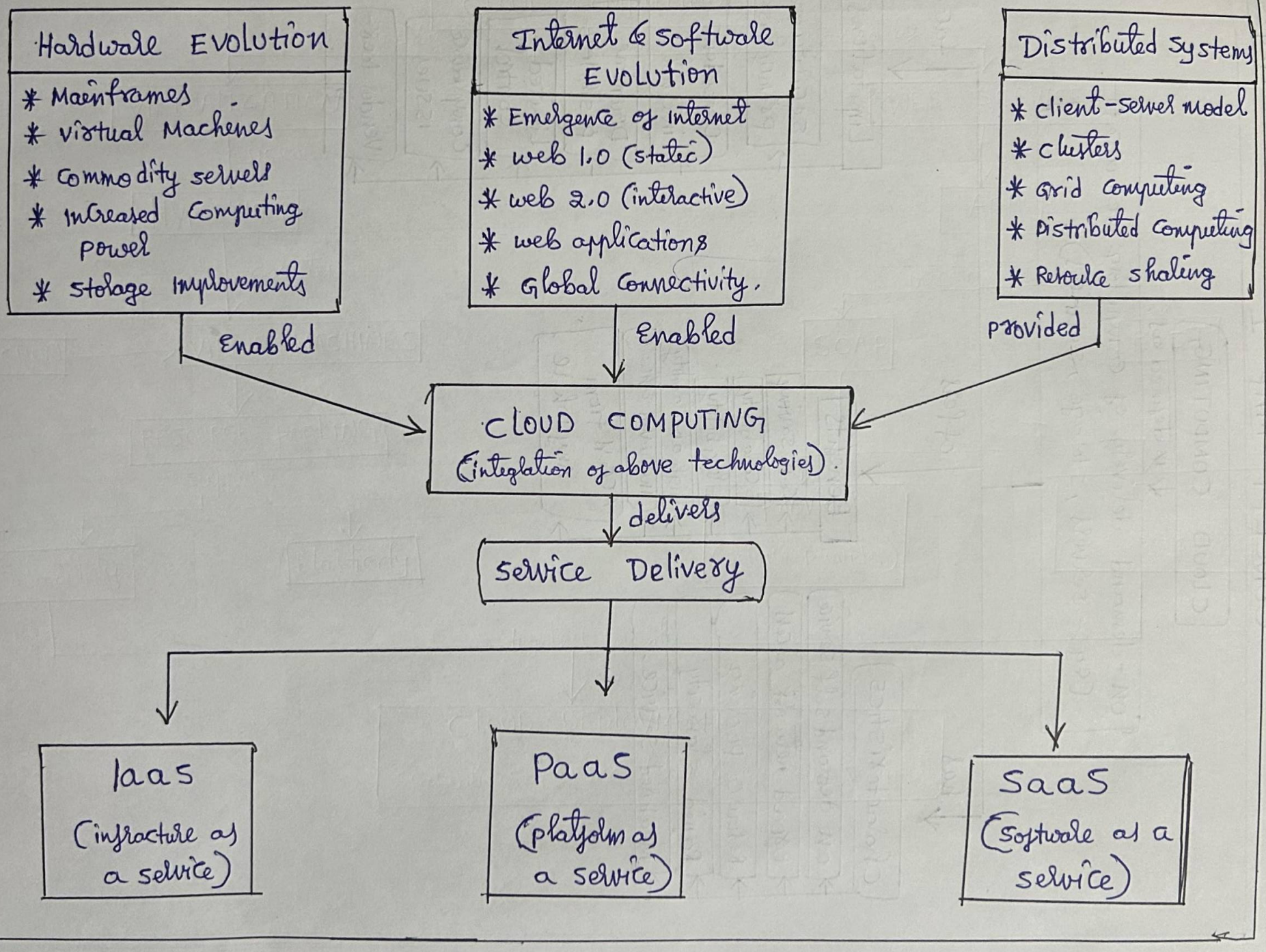
CLOUD COMPUTING FUNDAMENTALS

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CONCEPT MAP - 1



Concept Map 2:- EVOLUTION OF CLOUD COMPUTING



CONCEPT MAP-3 : CLOUD SERVICE MODELS

CLOUD SERVICE MODELS

include

IaaS (Infrastructure as a service)

provides

Virtualized infrastructure (servers, storage)

Provider Responsibility
* Hardware
* Networking
* Storage
* Virtualization.

User Responsibility
* Operating System
* Application
* Data
* Configuration.

PaaS (Platform as a service)

provides

Platform and development Environment (runtime, database, tools)

Provider Responsibility
* Infrastructure
* OS
* Runtime Environment
* Development tools.

User Responsibility
* Application
* Data
* User management

SaaS (Software as a service)

provides

Ready-to-use Software applications (email, CRM etc)

Provider Responsibility
* Infrastructure
* Platform
* Application
* Security & Maintenance

User Responsibility
* Data
* User settings
* Access management

XaaS (Anything as a service)

supports

Any IT service on demand (Eg: DBaaS, Faas, etc)

Provider Responsibility
* Service infrastructure
* Delivery & management
* Security
* Available.

User Responsibility
* Service configuration
* Usage
* Data
* Access Control.

Concept Map 4: Virtualization and web services as Enabler of Cloud.

